THE HENSELIZATION OF A PAIR OVER A NOETHERIAN RING IS NOETHERIAN

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. EDITORIAL NOTE: the statement contains a comment-like 'reference' environment (the chapter defines these environments with comment/endcomment), whose content includes a citation of EGA. Because that environment is not typeset, the citation is never processed and no bibliography entry is required; the stub that had been created for it was removed, and the retained preamble is now taken in full through its current last line so that the environment definition is present. The bytes of this note are plain ASCII and contain no control characters.

Allowed standard prerequisites (editorial).

- Henselization and strict henselization of a pair, with their universal properties. defined in the same section, source lines 10600-10977, before the counted statement. Supplies the rings whose Noetherianness is asserted.
- The strict henselization is a filtered colimit of finite etale algebras over the base. used at proof lines 11002-11030. Reduces the question to Noetherianness of finite type algebras.
- A filtered union of finitely generated subalgebras of a ring of finite type over a Noetherian base is Noetherian. used at proof lines 11030-11060. Is the specific statement that replaces the general false statement about filtered colimits.

Source reference. Original lemma 06LJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8468–8468. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 07QM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9837–9837. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 033E: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4053–4053. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 032B: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4014–4014. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00MC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 818–818. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 05UU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7528–7528. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 06LE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8463–8463. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07QN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9838–9838. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07ED: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9443–9443. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0C2Z: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15174–15174. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 04GM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5838–5838. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 04GU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5845–5845. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a local ring. The following are equivalent*

(1) *R is Noetherian,*
(2) *R^h is Noetherian, and*
(3) *R^{sh} is Noetherian.*

In this case we have

(a) $(R^h)^\wedge$ *and* $(R^{sh})^\wedge$ *are Noetherian complete local rings,*
(b) $R^\wedge \to (R^h)^\wedge$ *is an isomorphism,*
(c) $R^h \to (R^h)^\wedge$ *and* $R^{sh} \to (R^{sh})^\wedge$ *are flat,*
(d) $R^\wedge \to (R^{sh})^\wedge$ *is formally smooth in the* $\mathfrak{m}_{(R^{sh})^\wedge}$*-adic topology,*
(e) $(R^\wedge)^{sh} = R^\wedge \otimes_{R^h} R^{sh}$*, and*
(f) $((R^\wedge)^{sh})^\wedge = (R^{sh})^\wedge$.

Proof. Since $R \to R^h \to R^{sh}$ are faithfully flat (Lemma 07QM), we see that R^h or R^{sh} being Noetherian implies that R is Noetherian, see Algebra, Lemma 033E. In the rest of the proof we assume R is Noetherian.

As $\mathfrak{m} \subset R$ is finitely generated it follows that $\mathfrak{m}^h = \mathfrak{m}R^h$ and $\mathfrak{m}^{sh} = \mathfrak{m}R^{sh}$ are finitely generated, see Lemma 07QM. Hence $(R^h)^\wedge$ and $(R^{sh})^\wedge$ are Noetherian by Algebra, Lemma 032B. This proves (a).

Note that (b) is immediate from Lemma 07QM. In particular we see that $(R^h)^\wedge$ is flat over R, see Algebra, Lemma 00MC.

Next, we show that $R^h \to (R^h)^\wedge$ is flat. Write $R^h = \operatorname{colim}_i R_i$ as a directed colimit of localizations of étale R-algebras. By Algebra, Lemma 05UU if $(R^h)^\wedge$ is flat over each R_i, then $R^h \to (R^h)^\wedge$ is flat. Note that $R^h = R_i^h$ (by construction). Hence $R_i^\wedge = (R^h)^\wedge$ by part (b) is flat over R_i as desired. To finish the proof of (c) we show that $R^{sh} \to (R^{sh})^\wedge$ is flat. To do this, by a limit argument as above, it suffices to show that $(R^{sh})^\wedge$ is flat over R. Note that it follows from Lemma 07QM that $(R^{sh})^\wedge$ is the completion of a free R-module. By Lemma 06LE we see this is flat over R as desired. This finishes the proof of (c).

At this point we know (c) is true and that $(R^h)^\wedge$ and $(R^{sh})^\wedge$ are Noetherian. It follows from Algebra, Lemma 033E that R^h and R^{sh} are Noetherian.

Part (d) follows from Lemma 07QN and Lemma 07ED.

Part (e) follows from Algebra, Lemma 0C2Z and the fact that $R^\wedge$ is henselian by Algebra, Lemma 04GM.

Proof of (f). Using (e) there is a map $R^{sh} \to (R^\wedge)^{sh}$ which induces a map $(R^{sh})^\wedge \to ((R^\wedge)^{sh})^\wedge$ upon completion. Using (e) there is a map $R^\wedge \to (R^{sh})^\wedge$. Since $(R^{sh})^\wedge$ is strictly henselian (see above) this map induces a map $(R^\wedge)^{sh} \to (R^{sh})^\wedge$ by Algebra, Lemma 04GU. Completing we obtain a map $((R^\wedge)^{sh})^\wedge \to (R^{sh})^\wedge$. We omit the verification that these two maps are mutually inverse. □